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To the Editorial Office
Algorithms (MDPI)

Manuscript: *DT-GSK: Dimension-Tiered Adaptive Configuration Selection and Deterministic Refinement for Gaining-Sharing Knowledge Optimization* — algorithms-4507562, revision 1
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Dear Editors,

We are pleased to submit the revised version of our manuscript, *DT-GSK: Dimension-Tiered Adaptive Configuration Selection and Deterministic Refinement for Gaining-Sharing Knowledge Optimization* (algorithms-4507562). All ten reviewer comments are addressed. Four called for new evidence; we answered them with five pre-registered experiments (34,191 optimizer runs in total), each with its design, statistical conventions and adverse-outcome wording committed to the public repository before any result existed. Accompanying this letter are the point-by-point response, a marked-up main manuscript, a marked-up Supplement, and a change register listing every changed passage as submitted against as revised. One change the marked copies cannot show is the title page itself: the retitle Reviewer 1 requested sits in the document preamble, which latexdiff records without displaying, so we flag it here.

The revision strengthened what holds. The deterministic final refinement earns its place at both dimensions where it is active; the family standing survives a matched-initial-population control, staying in the top two at every CEC2017 dimension; and the dimension-tier boundaries are insensitive to perturbation at $D = 10$ and $D = 50$ — exactly where the narrowed contribution claims them. DT-GSK attains the best descriptive family-rank aggregate on CEC2017 (2.48) and CEC2013 (2.80), with the paper stating plainly that Holm-corrected tests separate it from the strongest baseline only at $D = 10$.

Three findings went against our submitted claims, and all three are in the paper rather than only in this letter. The learned eigenframe basis is outperformed by plain coordinate axes at both active dimensions under the manuscript's stated tie rule, so the refinement contribution is now claimed basis-neutrally and the eigenbasis is reported as a controlled negative result. The shipped $20 \leq D < 50$ profile is beaten from both neighbouring tiers at $D = 30$, extending a weakness the submitted manuscript had disclosed descriptively. And fixing the initial population at the comparators' value costs DT-GSK its first place at $D = 50$ and $D = 100$, so those two rank claims are qualified as resting in part on the population rule. We believe reporting these outcomes plainly makes the paper more useful, not weaker.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have read and approved the submitted version. For transparency, we note that A.W.M. is the originator of the baseline GSK algorithm and a co-author of several of the family variants used as comparators, one of which (eGSK) is also co-authored by H.S.M.R.; these relationships are declared in the manuscript's conflicts-of-interest statement. During the preparation and revision of this manuscript, the authors used two generative-AI assistants, Claude (Opus 4.6, 4.8 and 5.0; Anthropic) and ChatGPT (OpenAI), for language editing and rephrasing, for the drafting of expository prose restating findings the authors had already established from the frozen evidence, for structural review, consistency checking and review of the statistical and methodological descriptions, and for software-engineering support during implementation and tooling work, in accordance with the MDPI policy on the use of generative artificial intelligence. The algorithm design and the experimental protocol are the authors' own, and every reported number was produced by the authors' deterministic analysis pipeline from a version-locked evidence archive: no AI system designed an experiment, produced data, computed a statistic, or generated a scientific claim, result, or conclusion, and no AI system is an author of this work. The authors have reviewed, verified, and edited all AI-assisted text and take full responsibility for the content of this publication.

Sincerely,

Mostafa Elsayed Masoud (corresponding author)
Heba Sayed Mohamed Roshdy
Ali Wagdy Mohamed